Introduction to Ab Initio Nuclear Theory

Lecture 5: **Beyond the Slater Determinant: pairing correlations**

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Contents

1 Introduction to pairing correlations

In the previous Lecture we described how one goes from one- to many-body correlations, both in the formalism and in actual calculations, at least in principle. Reversely, we demonstrated how to capture many-body correlations in one-body correlations and how to go beyond. In this Lecture we will discuss concepts where this narrative gets complicated to the extent that it breaks practically breaks down. We will discuss pairing correlations as the simplest type of correlations that are yet elusive to capture with Slater Determinants in practice. We will start with a demonstration of the origin of pairing correlations, the Cooper instability. With the intuition acquired we will explore a mean-field theory that attempts to describe the many-body consequences of the Cooper instability, the BCS theory. As it will turn out, this will be a simple generalization of the Hartree-Fock (HF) theory for extensive systems in the presence of pairing correlations. Finally, we will generalize the BCS to the Hartree-Fock-Bogolyubov (HFB) theory for finite systems, which will bee the full generalization of HF theory for finite systems with pairing correlations.

2 The Cooper instability

The Cooper problem answers the question "what is the bound state of two particles in a Fermi gas, under attractive two-body interactions?".

Consider two particles with positions $\mathbf{r}_1$, $\mathbf{r}_2$ and spins σ_1,σ_2. At the simplest case, we assume a spin-independent interaction, $V(|\mathbf{r}_1 - \mathbf{r}_2|)$ which allows the two-body wavefunction to factor as $\psi_{\sigma_1\sigma_2}(\mathbf{r}_1, \mathbf{r}_2) = \varphi(\mathbf{r}_1, \mathbf{r}_2)\eta_{\sigma_1\sigma_2}$. Keep in mind that this doesn't hold for spin-dependent interactions, such as a spin-orbit coupling. Since the interaction only cares about the relative separation of the particles, i.e., there is no one-body forces, it is natural to decouple the centre-of-mass motion by transforming to Jacobi coordinates:

$$\mathbf{R} = \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2) \quad \Rightarrow \quad \mathbf{K} = \mathbf{k}_1 + \mathbf{k}_2 \tag{1}$$

$$\mathbf{r} = \frac{1}{2}(\mathbf{r}_1 - \mathbf{r}_2) \quad \Rightarrow \quad \mathbf{k} = \frac{1}{2}(\mathbf{k}_1 - \mathbf{k}_2) \tag{2}$$

where we also list the corresponding momenta for later use. The two-body wavefunction now is

$$\psi_{\sigma_1\sigma_2}(\mathbf{r}_1, \mathbf{r}_2) = \Phi(\mathbf{R})\phi(\mathbf{r})\eta_{\sigma_1\sigma_2} \tag{3}$$

From now on we will switch to a momentum-space formalism as it will turn out that pairs with zero centre-of-mass momentum are the most important. Schrödinger's equation momentum space now reads

$$\left(\hbar^2/m\mathbf{K}^2 + \hbar^2/m\mathbf{k}^2 - E\right)\varphi(\mathbf{K}, \mathbf{k}) = -{\sum_{\mathbf{k}'}}' V_{\mathbf{k}\mathbf{k}'}\varphi(\mathbf{K}, \mathbf{k}') \tag{4}$$

where the primed sum signifies summation of all $\mathbf{k}$-states above $\mathbf{k}_F$. This distinction is important: in the absence of the Fermi-sea ($k_F = 0$), Eq. (4)would yield a two-body bound state only if the potential $V$was strong enough, that is, only if the scattering length of V is negative. We will see that this is not the case for the filled Fermi-sea, $k_F \neq 0$. Focusing on $\mathbf{K} = 0$, when the number of available scattering states is maximum, we look for a in-medium bound state, i.e., a state with $E < 2E_F$. Expanding the potential in odd and even parts,

$$V_{\mathbf{k}\mathbf{k}'} = \sum_l (2l+1)V_l(k, k')P_l[\cos(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')] \tag{5}$$

and projecting Schrödinger's equation on a given partial wave, i.e., integrating it with a spherical harmonic Y_{lm}, we get

$$\left(\hbar^2/mk^2 - E\right)\phi_l(k) = -\frac{1}{2\pi^2}\int dk' k'^2 V_l(k, k')\phi_l(k') \tag{6}$$

Traditionally, one approximates $V_l(k,k')$ by a constant interaction $-V_l < 0$ in a $\mathbf{k}$-shell, i.e.,

$$V_l(k,k') = -V_l \text{ , for } k,k' \in [k_F, k_F + 2m\omega_c/\hbar] \tag{7}$$

$$= 0 \text{ ,} \quad \text{otherwise} \tag{8}$$

which yields a manifestly bound-state with energy $E_l = 2E_F - 2\hbar\omega_C\left[e^{\frac{4}{N(E_F)V_l}} - 1\right]^{-1}$. The dependence of E on V_l demonstrates that Cooper-pairing is not a perturbative phenomenon. This already hints towards what was alluded to in the introduction: pairing correlations are hard to capture perturbatively.

This was the two-body problem showing that the Fermi-sea is unstable under the formation of Cooper pairs. The natural many-body generalization of this is a condensation of such pairs which is described by the BCS wavefunction.

3 Bardeen Cooper Schrieffer theory

This section will serve as an overview of the formulation of the BCS theory to describe the s-wave superfluidity of homogeneous gases of neutrons and cold atoms. This is the common-textbook approach to superfluidity that will serve as guidance for later chapters. The BCS theory describes a fermionic superfluid as a condensation of pairs. It is important to note that in principle this is distinct from a condensation of dimers and the difference lays in the two-body system in free space. In free-space, two particles can form a dimer if their (attractive) interparticle interaction is strong enough to form a bound state. In the language of chapter **??**, this corresponds to an attractive interaction with a positive scattering length. In-medium, i.e., in the presence of other particles, two-particles can form a "bound" Cooper pair with an interaction too weak to form a free-space bound-state. In the language of **??**, this corresponds to an interaction with a negative scattering length. In what follows, we'll see this in detail, and we will revisit the subject of dimers at the end of this chapter.

3.1 The BCS ground state

In this overview of the BCS ground-state we will follow the way that will lead to later results most naturally. First we note that a singlet Cooper pair's wavefunction in second quantization reads:

$$|\psi\rangle = \sum_{\mathbf{k}}{}' \alpha(\mathbf{k}) c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} |F\rangle \tag{9}$$

where $|F\rangle = \prod_{\mathbf{k}}^{\mathbf{k}_F} c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{\mathbf{k}\downarrow} |0\rangle$, is the Fermi sea. Hence, a state where each $\mathbf{k}$ state is empty unless it contains a pair is

$$|\psi\rangle = \prod_{\mathbf{k}} \left[u_{\mathbf{k}} + v_{\mathbf{k}} c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow}\right] |0\rangle \tag{10}$$

where $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ are real variational parameters corresponding to probability amplitudes for the existence or non-existence of a pair on state $\mathbf{k}$. As such we have the nomralization condition $v^2_{\mathbf{k}} + u^2_{\mathbf{k}} = 1$. The BCS treatment amounts to minimizing the energy of the BCS state ignoring all normal-state interactions, and only considering the interaction responsible for the creation of the pairs which gives rise to the BCS Hamiltonian

$$H = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} c^\dagger_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma} + \sum_{\mathbf{kl}} V_{\mathbf{kl}} c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow} c^\dagger_{-\mathbf{l}\downarrow} c^\dagger_{\mathbf{l}\uparrow} \ . \tag{11}$$

whose expectation value is

$$\langle H\rangle = \sum_{\mathbf{k}} 2\epsilon_{\mathbf{k}} v^2_{\mathbf{k}} + \sum_{\mathbf{kl}} V_{\mathbf{kl}} v_{\mathbf{k}} u_{\mathbf{k}} v_{\mathbf{l}} u_{\mathbf{l}} \tag{12}$$

Expanding the product in Eq. (10) one can easily verify that the BCS wavefunction is not an eigenstate of the number operator, but rather describes an ensemble of systems

$$|\psi\rangle = \sum_N \lambda_N |\psi_N\rangle \tag{13}$$

where $|\psi_N\rangle$ is a state of N particles, defined via a projection operator $\mathcal{P}_N$: $|\psi_N\rangle = \mathcal{P}_N |\psi\rangle$. Practically, this means that when minimizing the energy one has make sure that the average particle number is kept constant which is achieved by introducing a Langrange multiplier, i.e., μ, the chemical potential. This amounts to minimizng the free energy of the system:

$$\langle F\rangle = \langle H\rangle - \mu \langle N\rangle = \sum_{\mathbf{k}} 2\xi_{\mathbf{k}} v_{\mathbf{k}}^2 + \sum_{\mathbf{kl}} V_{\mathbf{kl}} v_{\mathbf{k}} u_{\mathbf{k}} v_{\mathbf{l}} u_{\mathbf{l}} \; . \tag{14}$$

where $\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} - \mu$. We can now determine $v_{\mathbf{k}}$ and $u_{\mathbf{k}}$ by minimizing $\langle F\rangle$ subject to the normalization condition for $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$. This can be achieved straightforwardly by expressing

$$u_{\mathbf{k}} = \sin\theta_{\mathbf{k}} \quad \text{and} \quad v_{\mathbf{k}} = \cos\theta_{\mathbf{k}} \tag{15}$$

enforcing the normalization $u_{\mathbf{k}}^2 + v_{\mathbf{k}}^2 = 1$. In terms of these the Free energy becomes

$$\langle F\rangle [\theta_{\mathbf{k}}] = \sum_{\mathbf{k}} \xi_{\mathbf{k}}(1 + 2\cos\theta_{\mathbf{k}}) + \frac{1}{4}\sum_{\mathbf{kl}} V_{\mathbf{kl}} \sin 2\theta_{\mathbf{k}} \sin 2\theta_{\mathbf{l}} \; . \tag{16}$$

Minimizing this quantity amounts to requiring that its first derivative with respect to $\theta_{\mathbf{k}}$ vanishes (and the second derivative is positive; but we'll get to that later). The derivative reads,

$$\partial_{\theta_{\mathbf{k}}} \langle F\rangle [\theta_{\mathbf{k}}] = -2\xi_{\mathbf{k}} \sin 2\theta_{\mathbf{k}} + \sum_{\mathbf{l}} V_{\mathbf{kl}} \cos 2\theta_{\mathbf{k}} \sin 2\theta_{\mathbf{k}} \tag{17}$$

and setting it to 0 yields an equation for $\theta_{\mathbf{k}}$.

$$\tan 2\theta_{\mathbf{k}} = \frac{\sum_{\mathbf{l}} V_{\mathbf{kl}} \sin 2\theta_{\mathbf{k}}}{2\xi_{\mathbf{k}}} \tag{18}$$

We now define the gap function,

$$\Delta_{\mathbf{k}} = \langle\psi| \sum_{\mathbf{l}} V_{\mathbf{kl}} c_{\mathbf{l}\uparrow}^\dagger c_{-\mathbf{l}\downarrow}^\dagger |\psi\rangle = -\sum_{\mathbf{l}} V_{\mathbf{kl}} u_{\mathbf{l}} v_{\mathbf{l}} = -\frac{1}{2}\sum_{\mathbf{l}} V_{\mathbf{kl}} \sin 2\theta_{\mathbf{l}} \tag{19}$$

yielding

$$\tan 2\theta_{\mathbf{k}} = -\frac{\Delta_{\mathbf{k}}}{\xi_{\mathbf{k}}} \; , \quad \sin 2\theta_{\mathbf{k}} = \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}} \; , \quad \cos 2\theta_{\mathbf{k}} = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \; . \tag{20}$$

where we have also defined the quasiparticle excitation energy,

$$E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + \Delta_{\mathbf{k}}^2} \tag{21}$$

Knowing that $2u_{\mathbf{k}}v_{\mathbf{k}} = \sin 2\theta_{\mathbf{k}}$ and $v_{\mathbf{k}}^2 - u_{\mathbf{k}}^2 = \cos 2\theta_{\mathbf{k}}$, the conditions of Eq. (20) are enough to determine $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ in terms of the still unknown gap function $\Delta_{\mathbf{k}}$,

$$v_{\mathbf{k}}^2 = \frac{1}{2}\left(1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right) \; , \tag{22}$$

$$u_{\mathbf{k}}^2 = \frac{1}{2}\left(1 + \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right) \; , \tag{23}$$

Note how $u_{\mathbf{k}}^2 = 1 - v_{\mathbf{k}}^2$, as per their normalization. Plugging these into the definition of the gap function yields the *gap equations*

$$\Delta_{\mathbf{k}} = -\frac{1}{2}\sum_{\mathbf{l}} V_{\mathbf{kl}} \frac{\Delta_{\mathbf{l}}}{E_{\mathbf{l}}} \tag{24}$$

The chemical potential (Lagrange multiplier) provides the constraint on the particle-number equation, often called, the second gap equation,

$$\langle N \rangle = \sum_{\mathbf{k}} \left(1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right) \tag{25}$$

Interpreting the gap function as the binding energy of a pair with momentum $\mathbf{k}$ and when it's zero the BCS ground-state reduces to a Fermi sea. When $\Delta_{\mathbf{k}}$ is finite, it gives rise to a gapped quasiparticle excitation spectrum $E_{\mathbf{k}}$ from which we can define the order parameter of the superfluid, i.e., the pairing gap, as

$$\Delta = \min_{\mathbf{k}} E_{\mathbf{k}} \ . \tag{26}$$

Solving the gap equation for strongly interacting systems is demanding and it will be the topic of the last Lecture (Computational Lab). For now, we will employ an approximation routinely employed in condensed matter systems,

$$V_{\mathbf{kl}} = -V \ \text{, for } k, k' \in [\mu, \mu + 2m\omega_c/\hbar] \tag{27}$$
$$= 0 \ , \quad \text{otherwise} \ , \tag{28}$$

This approximation is invalid for nuclear systems whose interaction is active and non-constant over a range of momenta, but it can be instructive as it turns the gap function into a constant,

$$\Delta_{\mathbf{k}} = \Delta \ \text{, for } k, k' \in [\mu, \mu + 2m\omega_c/\hbar] \tag{29}$$
$$= 0 \ , \quad \text{otherwise} \tag{30}$$

and allows one to solve the gap equation

$$\Delta = \frac{\hbar\omega_c}{\sinh[1/N(E_F)V]} \approx 2\hbar\omega_c e^{-\frac{1}{N(E_F)V}} \tag{31}$$

Note the similarity of this result to the energy of a single Cooper pair in the Cooper instability: the BCS wavefunction describes a condensation of Cooper pairs with binding energy Δ. In strongly interacting systems, the approximation of Eq. (28) doesn't hold, the gap function is $\mathbf{k}$-dependent, and the gap equation must be solved self-consistently. Then the definition of the gap in Eq. (26) should be used for the pairing gap. Further, in practice, similarly to the approach followed for Cooper's problem, in the presence of a two-body interaction that is active mainly on a single partial-wave channel, the gap equation must be expanded on a partial wave basis, yielding

$$\Delta_{lm}(k) = -\frac{1}{\pi}\int dk' k'^2 V_l(k,k') \frac{\Delta_l(k')}{E(k')} \tag{32}$$
$$\langle N \rangle = \int dk k^2 \left(1 - \frac{\xi_k}{E_k}\right) \tag{33}$$

where

$$V_l(k,k') = \int dr r^2 j_l(kr) V(r) j_l(k'r) \tag{34}$$

with j_l the l-th spherical Bessel function.

3.2 Pairing from Slater Determinants and the BCS Determinant

In this section we have defined the BCS wavefunction which turned out to be a condensation of Cooper pairs, a phenomenon that can't be arrived to perturbatively. This type of wavefunction is the go-to description of pairing correlations and the point of departure for more refined descriptions of superfluidity. We will present some of these descriptions below, but first let's demonstrate what we've said multiple times already, that it is hard to describe the BCS wavefunction (and hence pairing) with Slater Determinants. Starting from the BCs wavefunction in Eq.(10), refactoring $u_{\mathbf{k}}$'s:

$$|\psi\rangle = \mathcal{U}\prod_{\mathbf{k}}\left(1+\gamma_{\mathbf{k}}c^\dagger_{\mathbf{k}\uparrow}c^\dagger_{-\mathbf{k}\downarrow}\right)|0\rangle \ , \quad \text{with} \quad \mathcal{U}=\prod_{\mathbf{k}}u_{\mathbf{k}} \ , \quad \gamma_{\mathbf{k}}=\frac{v_{\mathbf{k}}}{u_{\mathbf{k}}} \ . \tag{35}$$

Expanding the product in $\mathbf{k}$ we get

$$|\psi\rangle = \mathcal{U}\left(1+\sum_{\mathbf{k}}\gamma_{\mathbf{k}}c^\dagger_{\mathbf{k}\uparrow}c^\dagger_{-\mathbf{k}\downarrow}+\sum_{\mathbf{k}_1\mathbf{k}_2}\gamma_{\mathbf{k}_1}\gamma_{\mathbf{k}_2}c^\dagger_{\mathbf{k}_1\uparrow}c^\dagger_{-\mathbf{k}_1\downarrow}c^\dagger_{\mathbf{k}_2\uparrow}c^\dagger_{-\mathbf{k}_2\downarrow}+\cdots\right)|0\rangle \ . \tag{36}$$

Each one of these terms is a sum of Slater Determinants making the entire BCS wavefunction a sum of *many* Slater Determinants. Of course, some of this bad scaling arises because the BCS wavefunction doesn't conserve particle number. Particle conserving versions of the BCS theory only restrict the number of sums that show up in Eq. (36, but each sum remains a sum of many Slater Determinants.

However, this is not only bad news. The form of BCS wavefunction can be factored into another type of determinant, the BCS determinant. Indeed, taking a single term off Eq. (36) describing N particles we get

$$|\Psi_N\rangle = \sum_{\mathbf{k}_1\mathbf{k}_2\cdots\mathbf{k}_{N/2}}\gamma_{\mathbf{k}_1}\gamma_{\mathbf{k}_2}\cdots\gamma_{\mathbf{k}_{N/2}}c^\dagger_{\mathbf{k}_1\uparrow}c^\dagger_{-\mathbf{k}_1\downarrow}c^\dagger_{\mathbf{k}_2\uparrow}c^\dagger_{-\mathbf{k}_2\downarrow}\cdots c^\dagger_{\mathbf{k}_{N/2}\uparrow}c^\dagger_{-\mathbf{k}_{N/2}\downarrow}|0\rangle \tag{37}$$

Projecting this state on coordinate space

$$\langle\mathbf{X}|\Psi_N\rangle = \left\langle\mathbf{X}^\uparrow\mathbf{X}^\downarrow|\Psi_N\right\rangle = \sum_{\mathbf{k}_1\mathbf{k}_2\cdots\mathbf{k}_{N/2}}\gamma_{\mathbf{k}_1}\gamma_{\mathbf{k}_2}\cdots\gamma_{\mathbf{k}_{N/2}}\left\langle\mathbf{X}^\uparrow\mathbf{X}^\downarrow\right|c^\dagger_{\mathbf{k}_1\uparrow}c^\dagger_{-\mathbf{k}_1\downarrow}c^\dagger_{\mathbf{k}_2\uparrow}c^\dagger_{-\mathbf{k}_2\downarrow}\cdots c^\dagger_{\mathbf{k}_{N/2}\uparrow}c^\dagger_{-\mathbf{k}_{N/2}\downarrow}|0\rangle \tag{38}$$

$$= \sum_{\mathbf{k}_1\mathbf{k}_2\cdots\mathbf{k}_{N/2}}\gamma_{\mathbf{k}_1}\gamma_{\mathbf{k}_2}\cdots\gamma_{\mathbf{k}_{N/2}}\left\langle\mathbf{X}^\uparrow\right|c^\dagger_{\mathbf{k}_1\uparrow}c^\dagger_{\mathbf{k}_2\uparrow}\cdots c^\dagger_{\mathbf{k}_{N/2}\uparrow}|0\rangle\left\langle\mathbf{X}^\downarrow\right|c^\dagger_{-\mathbf{k}_1\downarrow}c^\dagger_{-\mathbf{k}_2\downarrow}\cdots c^\dagger_{-\mathbf{k}_{N/2}\downarrow}|0\rangle \tag{39}$$

where he have adopted the notation $|\mathbf{X}\rangle = \left|\mathbf{X}^\uparrow\mathbf{X}^\downarrow\right\rangle = \left|\mathbf{X}^\uparrow\right\rangle\left|\mathbf{X}^\downarrow\right\rangle$ separating the coordinates of up-spin and down-spin particles.

$$\langle\mathbf{X}^\sigma|\,a^\dagger_{n_1\sigma}\cdots a^\dagger_{n_N\sigma}|0\rangle = \langle\mathbf{X}^\sigma|\prod_{j=1}^{N_\sigma}c^\dagger_{\mathbf{k}_j\sigma}|0\rangle = \prod_{i=1}^{N_\sigma}\langle\{\mathbf{r}_i^\sigma\}|\prod_{j=1}^{N_\sigma}a^\dagger_{\mathbf{k}_j\sigma}|0\rangle = \langle 0|\prod_{i=1}^{N_\sigma}\prod_{j=1}^{N_\sigma}\psi_\sigma(\mathbf{r}_i)a^\dagger_{\mathbf{k}_j\sigma}|0\rangle \tag{40}$$

$$= \frac{1}{\sqrt{N!}}\det_{ij}\left[\phi_{\mathbf{k}_j}(\mathbf{r}_i)\right] \tag{41}$$

where the last equality is evaluated using Wick's theorem and noting the contraction

$$\langle 0|\,\psi_\sigma(\mathbf{r})a_{\mathbf{k}\sigma}|0\rangle = \phi_{\mathbf{k}}(\mathbf{r}) \tag{42}$$

And so

$$\left\langle \mathbf{X}^{\uparrow}\mathbf{X}^{\downarrow}|\Psi\right\rangle = \sum_{\mathbf{k}_1\mathbf{k}_2\cdots\mathbf{k}_{N/2}} \gamma_{\mathbf{k}_1}\gamma_{\mathbf{k}_2}\cdots\gamma_{\mathbf{k}_{N/2}} \frac{1}{\sqrt{N!}} \det_{ij}\left[\phi_{\mathbf{k}_j}(\mathbf{r}_i^{\uparrow})\right] \frac{1}{\sqrt{N!}} \det_{i'j'}\left[\phi_{-\mathbf{k}_{j'}}(\mathbf{r}_{i'}^{\downarrow})\right] \tag{43}$$

$$= \sum_{\mathbf{k}_1\mathbf{k}_2\cdots\mathbf{k}_{N/2}} \gamma_{\mathbf{k}_1}\gamma_{\mathbf{k}_2}\cdots\gamma_{\mathbf{k}_{N/2}} \frac{1}{N!} \sum_{\sigma\sigma'\in\mathcal{P}(N)} \eta_\sigma\eta_{\sigma'} \prod_j \phi_{n_j}[\mathbf{r}_{\sigma(j)}^{\uparrow}] \prod_{j'} \phi_{-\mathbf{k}_{j'}}[\mathbf{r}_{\sigma'(j')}^{\downarrow}] \tag{44}$$

$$= \sum_{\mathbf{k}_1\mathbf{k}_2\cdots\mathbf{k}_{N/2}} \gamma_{\mathbf{k}_1}\gamma_{\mathbf{k}_2}\cdots\gamma_{\mathbf{k}_{N/2}} \frac{1}{N!} \sum_{\sigma\sigma'\in\mathcal{P}(N)} \eta_\sigma\eta_{\sigma'} \prod_j \phi_{\mathbf{k}_j}[\mathbf{r}_{\sigma(j)}^{\uparrow}]\phi_{-\mathbf{k}_j}[\mathbf{r}_{\sigma'(j)}^{\downarrow}] \tag{45}$$

$$= \sum_{\mathbf{k}_1\mathbf{k}_2\cdots\mathbf{k}_{N/2}} \gamma_{\mathbf{k}_1}\gamma_{\mathbf{k}_2}\cdots\gamma_{\mathbf{k}_{N/2}} \frac{1}{N!} \sum_{\sigma\sigma'\in\mathcal{P}(N)} \eta_\sigma\eta_{\sigma'} \prod_j \Phi_{\mathbf{k}_j}[\mathbf{r}_{\sigma(j)}^{\uparrow} - \mathbf{r}_{\sigma'(j)}^{\downarrow}] \tag{46}$$

$$= \frac{1}{N!} \sum_{\sigma\sigma'\in\mathcal{P}(N)} \eta_\sigma\eta_{\sigma'} \prod_j \sum_{\mathbf{k}_j} \gamma_{\mathbf{k}_j}\Phi_{\mathbf{k}_j}[\mathbf{r}_{\sigma(j)}^{\uparrow} - \mathbf{r}_{\sigma'(j)}^{\downarrow}] \tag{47}$$

$$= \frac{1}{N!} \sum_{\sigma\sigma'\in\mathcal{P}(N)} \eta_\sigma\eta_{\sigma'} \prod_j \Phi[\mathbf{r}_{\sigma(j)}^{\uparrow} - \mathbf{r}_{\sigma'(j)}^{\downarrow}]\ . \tag{48}$$

Now since $\mathcal{P}(N)$ is a finite group, for every $\sigma' \in \mathcal{P}(N)$ there must exist $\sigma'' \in \mathcal{P}(N)$ such that $\sigma' = \sigma'' \circ \sigma$ or equivalently $\sigma'(j) = \sigma''(\sigma(j))$ for all j. Then $\sigma'' = \sigma' \circ \sigma^{-1}$ and so $\eta_{\sigma''} = \eta_{\sigma'}\eta_{\sigma^{-1}} = \eta_{\sigma'}\eta_\sigma$. Putting all of this together, we can replace all instances of σ' with $\sigma'' \circ \sigma$, the sign $\eta_{\sigma'}$ with $\eta_{\sigma''}\eta_\sigma$, and the sum over σ' with a sum over σ''. We will still be summing over the permutation group $\mathcal{P}(N)$ but from starting from σ instead of the identity element.

$$\left\langle \mathbf{X}^{\uparrow}\mathbf{X}^{\downarrow}|\Psi\right\rangle = \frac{1}{N!} \sum_{\sigma\sigma''\in\mathcal{P}(N)} \eta_\sigma\eta_{\sigma''}\eta_\sigma \prod_j \Phi[\mathbf{r}_{\sigma(j)}^{\uparrow} - \mathbf{r}_{\sigma''(\sigma(j))}^{\downarrow}] \tag{49}$$

$$= \frac{1}{N!} \sum_{\sigma\sigma''\in\mathcal{P}(N)} \eta_{\sigma''} \prod_j \Phi[\mathbf{r}_{\sigma(j)}^{\uparrow} - \mathbf{r}_{\sigma''(\sigma(j))}^{\downarrow}] \tag{50}$$

$$= \frac{1}{N!} \sum_{\sigma\in\mathcal{P}(N)} \det_{j\sigma(j)} \left(\Phi[\mathbf{r}_{\sigma(j)}^{\uparrow} - \mathbf{r}_{\sigma(j)}^{\downarrow}]\right) \tag{51}$$

$$= \frac{1}{N!} \det_{ji} \left(\Phi[\mathbf{r}_j^{\uparrow} - \mathbf{r}_i^{\downarrow}]\right) \sum_{\sigma\in\mathcal{P}(N)} \tag{52}$$

$$= \det_{ji} \left(\Phi[\mathbf{r}_j^{\uparrow} - \mathbf{r}_i^{\downarrow}]\right) \tag{53}$$

where we have used that the determinant is an invariant of the permutation group and so for any $\sigma \in \mathcal{P}(N)$ it is

$$\det_{j\sigma(j)} \left(\Phi[\mathbf{r}_{\sigma(j)}^{\uparrow} - \mathbf{r}_{\sigma(j)}^{\downarrow}]\right) = \det_{ji} \left(\Phi[\mathbf{r}_j^{\uparrow} - \mathbf{r}_i^{\downarrow}]\right) \tag{54}$$

To be maximally explicit this means

$$\left\langle \mathbf{X}^{\uparrow}\mathbf{X}^{\downarrow}|\Psi\right\rangle = \det \begin{pmatrix} \Phi(\mathbf{r}_1, \mathbf{r}_{1'}) & \Phi(\mathbf{r}_1, \mathbf{r}_{2'}) & \cdots & \Phi(\mathbf{r}_1, \mathbf{r}_{N'/2}) \\ \Phi(\mathbf{r}_2, \mathbf{r}_{1'}) & \Phi(\mathbf{r}_2, \mathbf{r}_{2'}) & \cdots & \Phi(\mathbf{r}_2, \mathbf{r}_{N'/2}) \\ \vdots & \vdots & \ddots & \vdots \\ \Phi(\mathbf{r}_{N/2}, \mathbf{r}_{1'}) & \Phi(\mathbf{r}_{N/2}, \mathbf{r}_{2'}) & \cdots & \Phi(\mathbf{r}_{N/2}, \mathbf{r}_{N'/2}) \end{pmatrix} \tag{55}$$

where we have now switched the notation for future convenience: the primed coordinates are the positions of the spin-down particles and the explicit dependence of Φ on the relative coordinate of the pair has been made implicit. The latter is because this assumption, while true in our cases here, is not necessary. It's worth noting that the BCS determinant is often abbreviated to $\mathcal{A}[\Phi(\mathbf{r}_1, \mathbf{r}_2), \Phi(\mathbf{r}_3, \mathbf{r}_4), \ldots, \Phi(\mathbf{r}_{N-1}, \mathbf{r}_N))]$. The fact that the BCs state can be written as a determinant is very useful as we will see in the QMC lecture, because determinants can be calculated with almost no waste of labor and are thus very cheap. Note that in the absence of pairing the BCS determinant factorizes into two Slated Determinants. This is reiterating the comment made earlier that both the normal state and the superfluid state can be described by the BCS wavefunction, which is essential in the wavefunction been interpreted as a variational one.

4 Hartree Fock Bogolyubov theory

The BCS theory attempts to capture pairing correlations to lower the ground-state energy while still remaining within the single-particle model. That is, in BCS the relevant degrees of freedom are quasiparticles which, from the practical point of view, are still single-particle excitations. One could then ask the question: "how many of the relevant correlations, *i.e.*, those that lower the ground state energy, can we capture while remaining in such a single-particle model?" The answer to this question come from the HFB theory which generalized the Bogolyubov transformation to capture the maximal amount of correlations while still yielding free "single-particle" quasiparticles.

Within HFB the Bogolyubov quasiparticles are defined via the generalized Bogolyubov transformation:

$$\beta_k^\dagger = \sum_l U_{lk} c_l^\dagger + V_{lk} c_l \tag{56}$$

$$c_l^\dagger = \sum_k U_{lk}^* \beta_k^\dagger + V_{lk}^* \beta_k \ . \tag{57}$$

where c ($c^\dagger$) are the annihilation (creation) operators associated with a complete set of single-particle states, e.g., harmonic oscillator, Woods-Saxon, or Nilsson orbitals, etc. The matrices U and V uniquely define the quasiparticle operators and can be seen as variational parameters.

One then defines the HFB ground-state as the quasiparticle vacuum. This can be uniquely identified, because $\{c\}$ and $\{\beta\}$ are both orthogonal basis sets, as the state generated by the maximal product of quasiparticle annihilation operators on the particle vacuum,

$$|\mathrm{HFB}\rangle = \prod_k \beta_k |0\rangle \ . \tag{58}$$

Therefore, identifying the matrices U and V in Eqs. (56) and (57) is enough to define the HFB ground-state. These matrices will prove to be the solutions of a set of equations called the HFB or Bogolyubov-deGennes equations. Typically one starts with a many-body Hamiltonian in a second-quantization form

$$H = \sum_{lm} t_{lm} c_l^\dagger c_m + \frac{1}{4} \sum_{lmnp} \bar{v}_{lm,np} c_l^\dagger c_m^\dagger c_p c_n \tag{59}$$

The HFB wavefunction is just a generalization of the BCS wavefunction and as such it also violates particle-number conservation. We have to constrain our calculations to the desired *average* particle number using a Lagrange multiplier. That is we replace H with $H_c = H - \lambda N$. To simplify notation, we will use H to signify H_c. The HFB ground-state will be an approximation of the exact ground-state of this Hamiltonian which we'll find using Ritz's variational principle: the HFB ground state is the one satisfying

$$\delta \frac{\langle\Phi| H |\Phi\rangle}{\langle\Phi|\Phi\rangle} = 0 \tag{60}$$

For such a variational procedure to be valid one needs a way of exploring the space of states close (by some measure) to the HFB ground-state. For that one uses Thouless' theorem which states that, for any reference state $|\Phi\rangle$, any state non-orthogonal to it, can be written as

$$|\Phi'\rangle = \exp\left(\sum_{k<k'} Z_{kk'}\beta_k^\dagger \beta_{k'}^\dagger\right)|\Phi\rangle \ . \tag{61}$$

To be able to calculate the expectation value of H in Eq. (59) with $|\Phi'\rangle$, we need to express H in the quasiparticle representation where it reads,

$$H = H^0 + \sum_{kk'} H^{11}_{kk'}\beta_k^\dagger \beta_{k'} + \sum_{k<k'}\left(H^{20}_{kk'}\beta_k^\dagger \beta_{k'}^\dagger + \text{h.c.}\right) + H' \ . \tag{62}$$

where H' contains terms with three or more creation (annihilation) operators. Now we can calculate

$$\frac{\langle\Phi| H |\Phi'\rangle}{\langle\Phi'|\Phi'\rangle} = H^0 + \left(H^{20*}\ H^{20}\right)\begin{pmatrix} Z \\ Z^* \end{pmatrix} + \frac{1}{2}\left(Z\ Z^*\right)\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix}\begin{pmatrix} Z \\ Z^* \end{pmatrix} \tag{63}$$

where

$$H^0 = \langle\Phi| H |\Phi\rangle \ , \quad H^{20}_{kk'} = \langle\Phi| [\beta_{k'}\beta_k, H] |\Phi\rangle \tag{64}$$

$$A_{kk'll'} = \langle\Phi| \left[\beta_{k'}\beta_k, \left[H, \beta_l^\dagger \beta_{l'}^\dagger\right]\right] |\Phi\rangle \ , \quad B_{kk'll'} = \langle\Phi| [\beta_{k'}\beta_k, [H, \beta_l \beta_{l'}]] |\Phi\rangle \tag{65}$$

This is a quadratic approximation of the multidimensional energy surface in the vicinity of $|\Phi\rangle$. So if $|\Phi\rangle$ satisfies Eq. (60), the linear term should vanish, that is, $H^{20}_{kk'} = 0$. However this doesn't define the quasiparticle operators uniquely since any unitary transformation of $\{\beta\}$ that doesn't mix β's with $\beta^\dagger$'s will leave $H^{20}_{kk'}$ unchanged. We can use this freedom to diagonalize $H^{11}_{kk'}$. That is, the HFB quasiparticles are defined as the ones eliminating H^{20} and simultaneously diagonalizing H^{11}. Organizing these in a matrix we get

$$\begin{pmatrix} H^{11} & H^{20} \\ -H^{20*} & -H^{11*} \end{pmatrix} \ , \quad \text{or in the particle space} \quad \begin{pmatrix} h & \Delta \\ -\Delta^* & -h^* \end{pmatrix} \tag{66}$$

with

$$\Gamma_{ll'} = \sum_{qq'} \bar{v}_{lq'l'q}\rho_{qq'} \ , \quad \Delta_{ll'} = \frac{1}{2}\sum_{qq'} \bar{v}_{ll'qq'}\kappa_{qq'} \tag{67}$$

$$\rho_{ll'} = \langle\Phi| c_{l'}^\dagger c_l |\Phi\rangle \ , \quad \kappa_{ll'} = \langle\Phi| c_{l'} c_l |\Phi\rangle \tag{68}$$

$$h = t + \Gamma - \lambda \tag{69}$$

Hence, solving the HFB equations amounts to solving the diagonalization problem

$$\begin{pmatrix} h & \Delta \\ -\Delta^* & -h^* \end{pmatrix}\begin{pmatrix} U_k \\ V_k \end{pmatrix} = \begin{pmatrix} U_k \\ V_k \end{pmatrix} E_k \ . \tag{70}$$

This will uniquely define the quasiparticle operators which in turn cast the Hamiltonian in a diagonal form:

$$H = H^0 + \sum_k E_k \beta_k^\dagger \beta_k + H' \tag{71}$$

where H' contains terms with more combinations of β and $\beta^\dagger$ which can be ignored to first approximation. Finally, and after having solved the HFB equations, the equivalent of a BCS state can be defined as

$$\Phi = \text{pf}(U^\dagger V^\star)\exp\left[\frac{1}{2}(VU^{-1})^\star_{ij} c_i^\dagger c_j^\dagger\right]|0\rangle \ . \tag{72}$$